

One CSI-Derived Rate Cannot Serve Every Budget: Budget Overrun and the Age Horizon in 5G Link Adaptation

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Abstract—A CSI report drives one rate decision, but different consumers hold that decision to different error budgets. We measure the mismatch on an NR LDPC-based link-level simulation. A policy whose achieved BLER sits near an eMBB budget of 10^{-1} overruns a URLLC budget of 10^{-3} by a factor of 112, at every SNR from cell edge to cell centre. We then calibrate the link so fresh CSI meets the 10^{-1} budget and ask how long a report may be held. The answer is a few milliseconds at walking pace and a single transmission interval at moderate mobility, so periodic reporting alone meets the budget only at the frame minimum. All results come from sealed pre-registered records, including a superseded relaxed-threshold pilot recomputed here at the calibrated target.

Index Terms—CSI feedback, link adaptation, URLLC, outer-loop link adaptation, reliability, 5G NR.

I. INTRODUCTION

Adaptive modulation and coding turns a CSI report into a rate decision. The transmitter selects the highest MCS whose required SINR the report clears, and the selection is tuned to an error budget. The traffic behind one report does not share one budget. An eMBB flow tolerates a block error rate of 10^{-1} ; a URLLC flow tolerates 10^{-3} or less. The same decision, from the same report, on the same channel, is judged against both.

Two quantities stay fixed throughout. The first is the achieved BLER $\hat{B}$ of a selection policy. The second is the budget overrun $\Omega(\beta) = \hat{B}/\beta$, the achieved BLER in units of the consumer's budget. $\hat{B}$ is a property of the policy and the channel. Ω depends on who is asking.

Two results follow. The first is partly mechanical and we say so: a policy run near 10^{-1} must overrun 10^{-3} by about a hundredfold. The second is the paper's core. With the link calibrated so that fresh CSI meets the 10^{-1} budget, the age horizon, the longest a report can be held before the budget is lost, collapses to a few transmission intervals at 10 Hz Doppler and to a single interval above 50 Hz. An earlier exploration at a relaxed threshold suggested a coherence-proportional law; at the budget itself there is almost no horizon to scale. The corrections are old, outer-loop link adaptation above all, and our measurement says when they have stopped being optional.

II. METHOD

Substrate. We use an NR LDPC-based link-to-system simulation at 3.5 GHz with a 1 ms TTI. For each MCS of the

NR table (TS 38.214 Table 5.1.3.1-1; QPSK-64-QAM, code rates $\geq 1/5$) we measure a block-error curve $e_m(\gamma)$ with the Sionna 5G LDPC decoder [2] over AWGN, 400 decoded blocks per point on a -8 to 28 dB grid at 2 dB spacing, linearly interpolated. The time-domain link draws a per-TTI SNR trace from a 3GPP TR38.901 TDL-A channel (30 ns delay spread, mean SNR 12 dB) and resolves each transmission as $\text{NACK}(t) \sim \text{Bernoulli}(e_m(\gamma(t)))$. Traces are 6000 TTI for the budget study and 12 000 TTI for the age study. Two accuracy limits bound our claims. At 400 blocks per point, zero measured errors still leaves a binomial 95% upper bound near 7.5×10^{-3} , so the curves are trustworthy at 10^{-2} and above and any smaller value is an interpolation, not a count. The mapping also ignores fading within a TTI.

System model. Let $\gamma(t)$ be the true SNR in dB at TTI t and $\hat{g}(t)$ the estimate the transmitter holds. Fresh CSI reads $\hat{g}(t) = \gamma(t) + n(t)$ with $n \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 1$ dB. A report held with period P reads $\hat{g}(t) = \gamma(t_k) + n(t_k)$ with $t_k = P\lfloor t/P \rfloor$: the estimate carries the report noise of its issue time and then ages. The certificate selects, at reference target $\beta_0 = 10^{-1}$ with backoff Δ ,

$$m^*(\hat{g}) = \max\{m : \rho_m(\beta_0) + \Delta \leq \hat{g}\}, \quad (1)$$

where $\rho_m(\beta) = \min\{\gamma : e_m(\gamma) \leq \beta\}$. The achieved BLER is the empirical NACK rate for the age study and the expected rate $\frac{1}{T} \sum_t e_{m^*}(\gamma(t))$ for the budget study, whose 10^{-3} scale cannot be resolved by counting at $T=6000$.

Policies. The budget-aware policy raises the backoff to $\Delta = 4$ dB and combines $K=3$ diversity branches, independent TDL-A realizations standing in for frequency or spatial replicas, so a block fails only if all branches fail. A backoff alone cannot reach 10^{-3} , because a single Rayleigh branch spends about 4% of its time in fades too deep for any MCS. Independence makes the diversity a best case, and the claim rests on a product rather than a counted deep tail: the per-branch rates sit at the percent level, where the curves are reliable. The correction of §III selects $m^*(\hat{g}(t) + o(t))$ and updates the offset on HARQ feedback,

$$o(t+1) = o(t) + \delta_{\text{up}} \mathbf{1}[\text{ACK}(t)] - \delta_{\text{dn}} \mathbf{1}[\text{NACK}(t)], \quad (2)$$

with $\delta_{\text{up}} = 0.1$ dB and $\delta_{\text{dn}} = \delta_{\text{up}}(1-\beta)/\beta$, so the loop settles where the NACK rate equals β .

Discipline. Each experiment is pre-registered with bars fixed before the graded run. Records are content-hashed, graded on seeds disjoint from design, and kept even when failed, voided or superseded [7]. Review of an earlier draft found the age sweep had used a relaxed 0.15 threshold while the text claimed 0.10; that exploration is kept, and §III reports the calibrated recomputation.

III. RESULTS

The reliability budget. Consider one CSI estimate judged under two budgets. Over three graded seeds of 6000 TTI with fresh CSI, the naive policy achieves $\hat{B} = 0.110$ to 0.113 , near its eMBB budget. The URLLC consumer reads the same value as an overrun of $112\times$. Part of this is arithmetic and we do not claim otherwise. The measurement adds the exact figure, the fact that nothing in the scheduler’s view flags it, and a feasible remedy: the budget-aware policy reaches 1.7×10^{-5} , 1.9×10^{-5} and below 10^{-5} on the three seeds, meeting the 10^{-3} budget with margin. That protection is explicitly specified, a 4 dB backoff and threefold diversity, though we do not quantify its spectral-efficiency cost here. The overrun is not an artifact of the operating point. A separately sealed cell sweeping mean SNR from 3 to 21 dB holds the URLLC overrun between $77\times$ and $142\times$ at every in-range SNR, and still $54\times$ where the MCS table saturates. Below about 6 dB even the eMBB consumer loses its budget, reaching $\hat{B} \approx 0.33$ at 3 dB, because the lowest MCS is already too fast for the channel. No margin repairs a certificate whose weakest format is too aggressive.

The age horizon. At 200 Hz Doppler with a 20-TTI report period the held estimate is off by 5.7 dB on average, aging dominating the 1 dB of report noise, and the naive certificate runs at $\hat{B} \approx 0.36$ against the 0.10 budget. OLLA corrects the same held report and holds the budget, 0.34 to 0.37 naive against 0.101 to 0.102 corrected over three sealed seeds.

How long may a report be held? Because the naive fresh policy lands just above the budget, the age study first calibrates: a fixed backoff $\Delta_{\text{cal}} = 0.5$ dB, chosen in 0.25 dB steps on a disclosed calibration seed, brings the fresh policy to $\hat{B} \leq 0.09$ at every Doppler, and all policies then carry it. The horizon is $P^*(f_D) = \max\{P : \hat{B}(P) \leq 0.10\}$, and a Doppler at which no period qualifies is recorded as censored rather than clamped. An exploratory sweep at a relaxed 0.15 threshold had suggested $P^* \approx 0.18 T_{\text{coh}}$. It was superseded by the calibrated experiment reported here, and both records are kept. At the true budget the answer is starker (Fig. 1). Over the three graded seeds P^* is 4, 6 and 4 TTI at 10 Hz, 2, 2 and 3 at 25 Hz, and 1 TTI, the frame minimum, at 50 Hz and above on every seed. The largest fresh-CSI $\hat{B}$ over all Dopplers is 0.078 to 0.080, confirming the calibrated baseline meets the budget, and no Doppler was censored. There is no room for a proportionality law because there is almost no horizon to scale. Once the budget is tight, the first few milliseconds of aging consume it. At 3.5 GHz these are modest speeds, about 3 km/h at 10 Hz and 15 km/h at 50 Hz. Below walking pace a report survives a few TTIs. Above it, periodic reporting meets the budget only at the frame minimum, so HARQ-driven correction is necessary

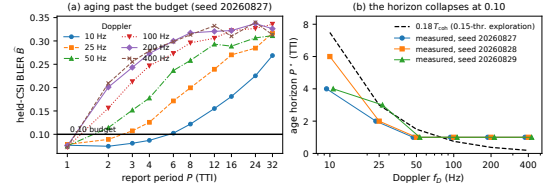


Fig. 1. The age horizon at the true 0.10 budget (calibrated baseline, three sealed graded seeds). (a) Held-CSI BLER against report period; lower is better. (b) The horizon collapses to a few TTI at low mobility and to the frame minimum above 50 Hz.

for any less aggressive cadence. The coherence-proportional behaviour is real, but it belongs to relaxed thresholds, where degradation has room to accumulate before the budget is lost.

IV. RELATED WORK AND CONCLUSION

Link adaptation, CSI aging and outer-loop correction are old subjects. OLLA [1] holds an error target under imperfect or stale CSI by adjusting an SINR offset on HARQ feedback, and it is exactly the correction our age result makes mandatory. Short-packet analysis [3] establishes the reliability regime where the budget gap lives. NR configures CSI reporting as periodic, semi-persistent or aperiodic [4]; none adapt the cadence to the consumer’s budget, and our horizon result says how little cadence alone can do at a tight one. Event-based sampling [5] we tested only in exploratory form, and a sealed confirmation attempt was voided by a control check, so we claim nothing there. For learned CSI feedback [6], the objective gap it might suggest did not appear in our tests at practical operating points. We offer no new estimator, scheduler or offset rule. We offer the measurement.

The unit of correctness in link adaptation is not the CSI report. It is the pair of report and budget. Sized to the consumer’s budget, margins and reporting cadence keep the certificate accurate; sized to a nominal channel they do not, and the costs are now numbers. The limits are worth stating: one link-to-system simulation, one TDL-A profile, one MCS table topping at 64-QAM, and values below 10^{-3} limited by curve resolution. Over-the-air HARQ traces are the natural next step.

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